4.4 Factoring Polynomials

Name _____ Period ____ Date _____

Algebra II 4.4 · Student sheet

I can…

- factor polynomials by factoring out a monomial and by grouping.
- factor sums and differences of cubes and polynomials in quadratic form.
- use the Factor Theorem.

Vocabulary · fill in the missing word

Factored completely — written as a product of _____ polynomials with integer coefficients

Sum of two cubes — $a^3 + b^3$ = _____

Difference of two cubes — $a^3 - b^3$ = _____

Quadratic form — $au^2 + bu + c$, where u is an _____ such as x^2

Factor Theorem — $x - k$ is a factor of f(x) _____ when f(k) = 0

Part 1 · Factor Theorem hunt

$f(x) = x^3 - 2x^2 - 5x + 6$

x	−3	−2	−1	0	1	2	3
f(x)							

Zeros: _____ Factored form: f(x) = _____

Part 2 · Notes: fill in the blanks

$a^3 + b^3$ = _____

$a^3 - b^3$ = _____

Factor Theorem: $x - k$ is a factor exactly when _____.

Part 3 · Examples (work them with the class)

Example 1

Factor completely: $2x^4 - 8x^2$

Example 2

Factor completely.

a. $x^3 + 27$ b. $8x^3 - 1$

Example 3

Factor completely: $x^3 - 3x^2 - 4x + 12$

Example 4

Factor completely: $x^4 - 5x^2 + 4$

Example 5

Given that $x - 4$ is a factor of $f(x) = x^3 - 6x^2 + 5x + 12$, factor $f(x)$ completely.

Part 4 · Partner practice

Partner 1

Factor $x^3 - 64$.

Partner 2

Is $x + 1$ a factor of $f(x) = x^3 + 2x^2 - x - 2$?